

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1712

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Devadharshini M(192572149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

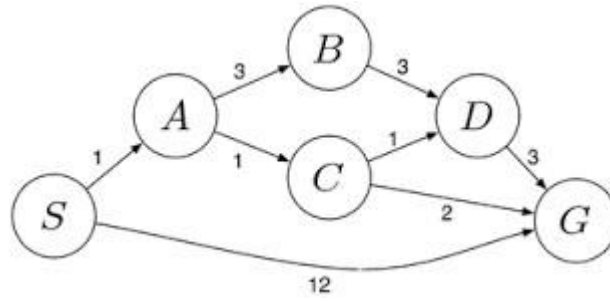
Signature of the student: Devadharshini M

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			
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1. Water Jug Problem (4-gallon and 3-gallon jugs)

Initial State: (0,0)

Goal State: (2,0) → 2 gallons in the 4-gallon jug.

Step	4-Gallon Jug	3-Gallon Jug	Action
1	0	3	Fill 3-gallon jug
2	3	0	Pour into 4-gallon jug
3	3	3	Fill 3-gallon jug again
4	4	2	Pour into 4-gallon until full
5	0	2	Empty 4-gallon jug
6	2	0	Pour remaining 2 gallons into 4-gallon jug

Goal Achieved: The 4-gallon jug contains **exactly 2 gallons**.

2. Mars Rover Agent

(i) Percepts

- Camera images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Terrain information
- Position (GPS/Inertial sensors)

(ii) Operating Environment

- **Partially Observable**
- **Sequential**
- **Dynamic**
- **Continuous**
- **Single-agent**

(iii) Actions

- Move forward/backward
- Turn left/right
- Capture images
- Collect rock samples
- Drill rocks
- Analyze samples
- Send data to Earth

(iv) Performance Measure

- Accurate sample collection
- Maximum area explored
- Safe navigation
- Efficient energy usage
- Successful transmission of data

(v) Suitable Agent Architecture

Utility-Based Learning Agent

Reason: It can evaluate different actions based on safety, energy consumption, and scientific value while improving its decisions from experience.

3. 8-Queens Problem Formulation

Initial State

Empty 8×8 chessboard.

State Space

All possible arrangements of 8 queens.

Actions

Place one queen in a valid position in the next column.

Goal Test

- Eight queens placed.
- No two queens share the same:
 - Row
 - Column
 - Diagonal

Path Cost

Each queen placement has equal cost.

Search Strategy

Backtracking Search / Depth-First Search (DFS)

Sample Solution

Column	:	1	2	3	4	5	6	7	8
Row	:	1	5	8	6	3	7	2	4

This arrangement satisfies all constraints.

4. OLA Cab Booking Problem

Problem-Solving Agent

Goal-Based Agent

Reason

The goal is to select the best cab type (Mini, Micro, Sedan, Prime, etc.) and reach the destination.

Pseudocode

START

Input pickup location

Input destination

Display available cab types

User selects cab

IF cab is available THEN

 Calculate fare

 Book cab

 Show driver details

 Navigate to destination

ELSE

 Display "Cab Not Available"

ENDIF

END

5. Uniform Cost Search (UCS)

- Given Edges:
- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

- UCS Steps

Step	Expanded Node	Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	G	4 (Goal Reached)

Optimal Path- $S \rightarrow A \rightarrow C \rightarrow G$

- Total Cost- $1 + 1 + 2 = 4$

Result

Using the Uniform Cost Search (UCS) algorithm, the least-cost path is $S \rightarrow A \rightarrow C \rightarrow G$ with a minimum cost of 4. UCS guarantees the optimal path by always expanding the node with the lowest cumulative cost.